

Dexter Fernandes

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Computer Vision Engineer

Computer Vision Engineer with 3+ years building real-time perception systems for industrial safety, edge hardware and harsh field environments, including hot, dusty and compute-constrained settings. Experienced across object detection, tracking, SLAM, sensor fusion and point cloud processing using C++17, Python, ROS2, OpenCV, PCL, Open3D and Linux. Strong deployment background with OpenVINO, FP16/INT8 quantisation, Intel VPU hardware and 24/7 perception pipelines where reliability, latency and failure visibility matter. MSc Robotics graduate with thesis work in SLAM and LiDAR perception for GPS-denied environments, with practical relevance to onboard scene understanding and autonomous systems.

WORK EXPERIENCE

Esbaa • Muscat, Oman

Jan 2023 - Aug 2023

AI Engineer

- Owned the end-to-end computer vision pipeline for real-time perception in harsh industrial environments, covering data curation, model training through deployment and on-site validation, improving F1 by 15% while maintaining deterministic 24/7 operation.
- Built multimodal data ingestion, cleansing, training, and validation pipelines across image, video, and sensor data using Python, Pandas, Airflow, MLflow, DVC, Optuna, and CVAT, processing 3.6 million records and reducing model rollout time from 3 days to under 24 hours.
- Led on-site validation and diagnostics on BP oil-rig deployments, troubleshooting model failures under environmental stressors and translating performance findings into fixes that improved robustness in live operation.
- Refactored legacy services into Dockerised microservices with Kafka-based messaging, structured logging, health checks, retries, and safe restarts, improving fault isolation, increasing MTBF by 35%, and reducing live video pipeline crashes by 67%.
- Built benchmarking and profiling harnesses to evaluate open-source models (YOLOv5, EfficientDet) and quantify accuracy, latency, and stability pre- and post-optimisation for regression detection.
- Documented deployment guardrails, failure modes, and recovery procedures, improving maintainability for engineers supporting fielded systems.

LivNSense Technologies • Bengaluru, India

May 2020 - Dec 2022

Lead Software Engineer - AI

- Led a team to design, train, and deploy production object detection and tracking models for industrial safety applications, reducing false positives by 33% while meeting latency and power constraints.
- Built low-latency perception pipelines using OpenVINO, and C++17, tripling throughput from 5 FPS to 15 FPS and delivering deterministic real-time inference on Intel VPU hardware under tight power and memory constraints.
- Deployed quantised vision models using FP16/INT8 optimisation, pruning and layer fusion, reducing model size by 50% while preserving production accuracy targets.
- Built automated labelling, dataset versioning, and performance-tracking workflows, repurposing existing detection models for annotation and cutting manual labelling effort by 80% while accelerating iteration cycles on large real-world datasets.
- Developed synthetic data generation and augmentation tools to improve edge-case coverage and cut manual labelling time by 60% using OpenCV and Albumentations.
- Integrated perception modules with hardware, software and QA teams, translating model performance, latency and failure analysis into rollout decisions for deployed safety systems.

PROJECTS

Modern Edge CV Benchmarking

Mar 2026

- Benchmarked modern computer vision models including YOLOv8, YOLOv10, RT-DETR, EfficientViT, Grounding DINO, MobileSAM, and Depth Anything v2 to compare accuracy, latency, model size, and deployment suitability.
- Evaluated Vision Transformer and foundation-model approaches for object detection, segmentation, depth estimation, and open-vocabulary visual understanding, with a focus on production-style CV trade-offs.
- Exported models to ONNX and evaluated OpenVINO deployment paths, strengthening practical understanding of model optimisation, inference acceleration, and edge deployment constraints.
- Built benchmark comparisons using mAP, inference time, and model-size measurements to support model selection for low-latency visual AI systems.

ROS2 SLAM & Nav2 Simulation

Feb 2026

- Built a simulation-based robotics stack in ROS2 Jazzy and Gazebo Harmonic to evaluate LiDAR-centric SLAM, LiDAR+RGB fusion SLAM, and autonomous navigation in structured scenarios.
- Benchmarked RTAB-Map, LIO-SAM, Cartographer, and GLIM across 2D and 3D LiDAR configurations, comparing mapping and localisation behaviour to support system selection and iteration.
- Developed ROS2 nodes, launch files, and parameterised configurations in C++ and Python, using tf2, message synchronisation, camera and sensor ingestion, and timestamp alignment to maintain frame consistency and debug transform timing issues.
- Implemented point cloud processing with PCL and Open3D for filtering, segmentation, and registration, and added an EKF-based localisation layer with robot_localization to fuse wheel odometry and IMU for smoother odom-to-base tracking during LiDAR-only SLAM.
- Containerised the stack with Docker on Linux and managed versioning in Git to support reproducible builds and consistent execution across development environments.

End-to-End Damage Severity & Claim Assessment

Nov 2025

- Developed an automotive claims assessment pipeline combining computer vision-based damage detection with severity classification to support faster triage of real-world vehicle damage scenarios.
- Built a multimodal image-to-structured-output workflow that converted vehicle-damage detections and severity predictions into insurer-friendly claim summaries using structured reasoning and LLM-generated explanations.
- Designed the full workflow across data preprocessing, model training, evaluation, and a user-facing inference module, demonstrating end-to-end delivery from model output to usable product insight.
- Created interpretable repair insights from CV model outputs, translating visual predictions into clear claim explanations for non-technical users.

EDUCATION

Master of Science in Robotics - Merit

University of Bristol • Bristol, England

Sep 2023 - Sep 2024

Dissertation: Radiological SLAM with LiDAR Odometry

- Radiological SLAM with LiDAR Odometry. Designed and implemented a real-time SLAM system using RTAB-Map on the Clearpath Husky platform, integrating LiDAR and radiological data for localisation and mapping in inspection scenarios.
- Evaluated RTAB-Map, LIO-SAM, and Cartographer across comparable scenarios using ROS2 and rosbag2-based workflows to assess mapping and localisation behaviour.
- Validated sensor placement for coverage and calibrated LiDAR extrinsics to the robot base, including the Velodyne VLP-16 and Intel RealSense D435i.
- Implemented a ROS2 (rclcpp) point cloud pipeline in C++17 using PCL, and developed ROS2 nodes with tf2, message synchronisation, and sensor data playback workflows to maintain frame consistency and debug transform timing issues.
- Worked with camera and sensor ingestion in ROS2 pipelines, including timestamp alignment and downstream processing for SLAM and perception tasks.

Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

Jul 2016 - May 2020

- Built a strong foundation in signals, systems, embedded hardware, and control, which later supported work in real-time AI, sensor systems, and robotics software.

SKILLS

- **SLAM, Camera and Data Pipelines:** Cartographer, Ceres, Eigen, Filtering, GLIM, LiDAR SLAM, LIO-SAM, Multi-camera perception, Point cloud processing, Pose graph optimisation, Registration, RTAB-Map, Scan matching, Segmentation, Video and sensor data processing pipelines, Visual data streams
- **Cloud, Edge & MLOps:** Apache Airflow, Apache Kafka, API design, AWS, CI/CD, Docker, DVC (Data Version Control), Health checks, Intel VPUs, Microsoft Azure, MLFlow, Post-deployment validation, Production monitoring, Structured logging
- **Programming & Tooling:** Bash, C++17, CMake, colcon, Django, FastAPI, Git, Intel VTune, Linux, NumPy, OpenCV, Pandas, PG vector, PIL, PostgreSQL, Python, Qt, Scikit-image, Scikit-Learn
- **Deep Learning and Optimisation:** Bayesian Hyperparameter Optimisation, Convolutional Neural Networks, FP16/INT8 deployment, Layer fusion, ONNX, OpenVINO, Optuna, Post-training Quantisation, PyTorch, Quantisation-Aware Training, Scikit-Learn, Structured pruning, TensorFlow, TensorRT, YOLO
- **Edge AI and Inference:** Intel Core i7 (VTC7252-7C4IP), Intel Movidius Myriad VPU, NVIDIA Jetson AGX Xavier
- **Professional Skills:** Agile framework, Experiment-driven development, Stakeholder communication, Technical leadership
- **Point Cloud & Vision Processing:** Denoising, Open3D, OpenCV, Point Cloud Library, Scikit-Image
- **Robotics and Middleware:** Gazebo Harmonic, Nav2, rclcpp, rclpy, Real-time perception systems, robot_localization, ROS2 (Robot Operating System 2), rosbag2, sensor data playback, Sensor integration, tf2, timestamp synchronisation
- **Computer Vision & Image Processing:** Anomaly detection, Data augmentation, Defect detection, Image processing, LiDAR extrinsics calibration, Object detection, Object tracking, Segmentation, Sensor calibration, Visual quality inspection
- **AI-Assisted Development:** ChatGPT Codex, Claude Code, Prompt engineering
- **Computer Vision Models:** ByteTrack, DeepLabv3+, Depth Anything v2, EfficientDet, EfficientViT, Faster R-CNN, Grounding DINO, Mask R-CNN, MobileSAM, RT-DETR, SegFormer, SSD MobileNet V2, U-Net, YOLOv10, YOLOv5, YOLOv8
- **Datasets, Annotation & Evaluation:** Automated labelling, Cityscapes, COCO, Confusion-matrix-driven failure clustering, CVAT, Dataset versioning, DVC, F1-score, ImageNet, IoU, KITTI, Latency benchmarking, mAP, Model-size benchmarking, PASCAL VOC, Precision/Recall, Synthetic data generation